Coordinates of RT-300S and FRED

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FRED format

- FRED format for Tabulated BSDF:
  - Best documentation/resource currently found: https://support.photonengr.com/article/304-viavi-engineered-diffuser-bsdf
- Format of txt file for FRED:
  - θ_i (0) φ_i (0)
    - θ_s (0) φ_s (0) BSDF value
    - θ_s (0) φ_s (1) BSDF value
    - θ_s (0) φ_s (2) BSDF value
    - …
    - θ_s (1) φ_s (0) BSDF value
    - θ_s (1) φ_s (1) BSDF value
    - …
    - θ_s (end) φ_s (end) BSDF value
  - θ_i (0) φ_i (1)
    - …
  - …
  - θ_i (end) φ_i (end)

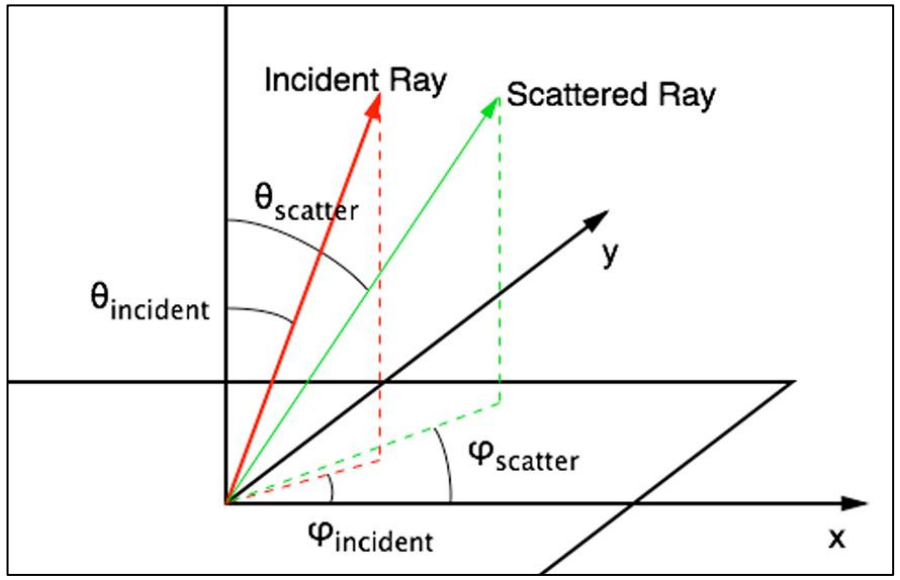


Fig 1. FRED angle definitions.

Angle definitions:

- $\theta_i \equiv$ polar incident
- $\varphi_i \equiv$ azimuth incident
- $\theta_s \equiv$ polar scatter
- $\varphi_s \equiv$ azimuth scatter

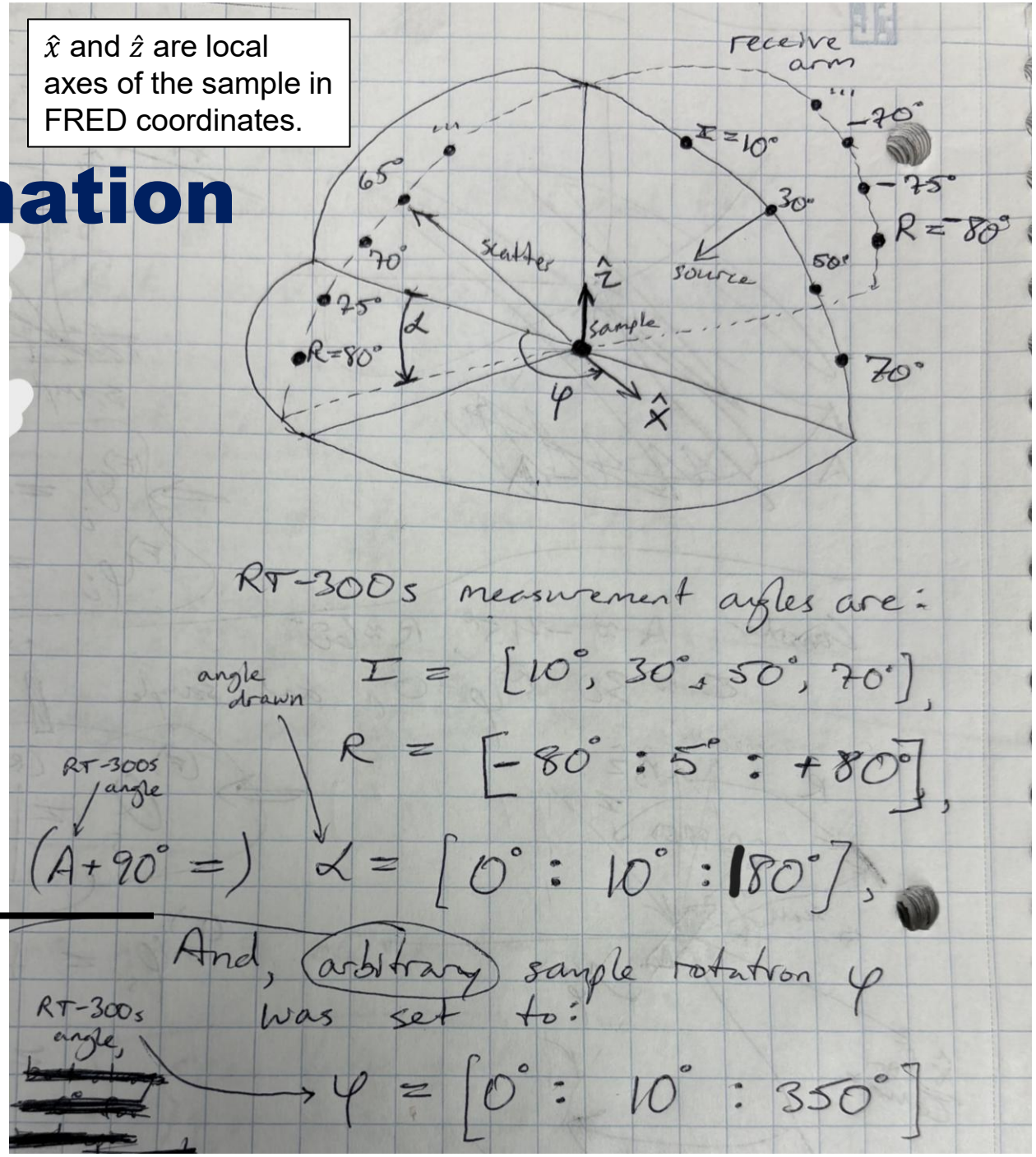
Coordinate transformation

- From 2026/01/21 LOFT notebook,

FRED	RT-300S (machine coordinates)
θ_i	I
* φ_i	$-\varphi$
θ_s	$\lvert R \rvert$
** φ_s	$90° + A - \varphi + \big(1 - \mathrm{sign}(R)\big)(90°)$, or all real numbers if $R = 0$

*φ is manually-set stage rotation;
**the (1-sign(R))(90) just adds 180 if R<0.

$\hat{x}$ and $\hat{z}$ are local axes of the sample in FRED coordinates.



Arbitrary because not actually measured, just hard-coded in FRED to manually force isotropic behavior – not sure if “isotropic setting” exists like in Zemax bsdf format